Galois Theory for Rats - Galois Theory 101

Skyler Hu

March 5, 2026

Contents

1 Introduction

After our prerequisites (Field Extensions, Field Representation & Construction), hopefully you have gained some basic ideas of field theory. If so, congratulations! You are officially a graduate of Galois Theory kindergarten and can now move on to the *real* Galois Theory. Warning: the math is going to get insidious from now on, so be prepared.

In this fourth handout of Galois Theory for Rats, we take the crucial step from fields to groups – the core of Galois Theory, and show how groups, fields and polynomials are closely interrelated. After some preliminary definitions, we get to enjoy the *Galois Correspondence* (or *The Fundamental Theorem of Galois Theory*), where everything we've learned will start to fall together and (ideally) you will understand why Galois Theory is considered one of the most beautiful branches of mathematics. Note that from now on, logic and proofs take up a significantly greater part of the handouts, and unlike the previous ones where proofs for most results and optional, you will be forced to understand proofs, some of them rather painful. You are encouraged to read what you don't understand as many times as you need to – this is the journey of every junior mathematician.

2 Fields' Groups and Groups' Fields

In this section we introduce some definitions central to Galois Theory. These definitions are mainly established through the process of creating groups from fields, fields from groups and both from polynomials. We start with another definition ending in '-morphism':

Definition 2.1. (Field Automorphism) An isomorphism σ of field K is called an *autormorphism* if it maps every element in K onto elements in K. The collection of automorphisms of a field K is denoted $Aut(K)$.

There exists many automorphisms of a field K since there are multiple ways to shuffle the elements of K around, and it follows naturally that we'd be interested in a special category of field automorphisms: these automorphisms map some elements onto themselves.

Definition 2.2. An automorphism $\sigma \in Aut(K)$ is said to *fix* an element $\alpha \in K$ if $\sigma\alpha = \alpha$. If F is a subset/subfield of K, then an automorphism of K is said to *fix* F if it fixes every element in F. $Aut(K/F)$ is defined as the collection of automorphisms of K fixing F.

Proposition 2.1. (Automorphisms form a group.) The collection of automorphisms of field K, $Aut(K)$, is a group under composition. It has subgroup $Aut(K/F)$ for F a subfield of K.

Proof. Can you see that this is obvious? (Hint: check group axioms) □

Now that we've seen some very basic properties of automorphisms, we're ready to address a common statement made about Galois Theory: the permutation of roots. Though at this point we still have a long way to go until we learn precisely how root permutation relates to solvability, right now we can show that automorphisms permute the roots of polynomials. More specifically:

Proposition 2.2. (Automorphism permutes roots.) Let K/F be a field extension and let α be algebraic over F. Then for any automorphism $\sigma \in Aut(K/F)$, $\sigma\alpha$ is a root of the minimal polynomial of α over F, i.e. $Aut(K/F)$ permutes the roots of irreducible polynomials, and any polynomial in $F[x]$ having α as a root also has $\sigma\alpha$ as a root.

Proof. (Short and Easy!) Let α be a root of the polynomial $f(x) \in F[x]$

$$f(x) = a_n x^n a_{n-1} x^{n-1} + \ldots + a_1 x + a_0.$$

Then $a_n\alpha^n + a_{n-1}\alpha^{n-1} + \ldots + a_1\alpha + a_0 = 0$. Applying σ to each term gives us

$$\sigma(a_n\alpha^n) + \sigma(a_{n-1}\alpha^{n-1}) + \ldots + \sigma(a_1\alpha) + \sigma(a_0) = \sigma(0).$$

□

Note. The *minimal polynomial* of α over F is defined to be the irreducible polynomial $m(x) \in F[x]$ of least degree having α as a root; in a sense, the most simple polynomial with α as a root. It is often algebraic shorthand to say that something (in this case $\sigma\alpha$) is the root of $m(x)$ instead of 'every polynomial having α as a root also has $\sigma\alpha$ as a root.'

Example 2.1.

1. Consider the field extension $\mathbb{Q}(\sqrt{2})$. If τ is an automorphism of $\mathbb{Q}(\sqrt{2})$ (i.e. $\tau \in Aut(\mathbb{Q}(\sqrt{2}))$), then $\tau(\sqrt{2}) = \pm\sqrt{2}$, which are both roots of the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$, which is $x^2 - 2$.
   Since τ fixes $\mathbb{Q}$, we know that τ is of the form
   $$\tau(a + b\sqrt{2}) = a \pm b\sqrt{2}.$$
   $\tau(\sqrt{2}) = \sqrt{2}$ is the identity automorphism, and $\tau(\sqrt{2}) = -\sqrt{2}$ has order 2, so the group $Aut(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$ is the cyclic group of order 2.
2. Consider the field extensions $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$. You might be tempted to think that the automorphism group with $\mathbb{Q}$ fixed is a cyclic group of order 3, but consider this: The minimal polynomial of $\sqrt[3]{2}$ is $x^3 - 2$, which has one root $\sqrt[3]{2}$ with multiplicity 3. Therefore the automorphism fixing $\mathbb{Q}$ also fixes $\sqrt[3]{2}$ (and fixes $\sqrt[3]{4}$ for that matter), and $Aut(\mathbb{Q}(\sqrt[3]{2})) = \{1\}/$

We have seen that field extensions can create groups by considering the automorphism group of a field F (usually fixing a subfield of F). Now, in order to fully appreciate the relationship between these algebraic structures, we will see how groups can create fields:

Proposition 2.3. (Field Extensions from Groups) Let $H \leq Aut(K)$ be a subgroup of $Aut(K)$. The collection F of elements in K that are fixed by all elements in H form a subfield of K.

Proof. Check field axioms. Be sure to use the definition of an isomorphism that $\tau(ab) = \tau(a)\tau(b)$. □

Proposition 2.3 gives us a field from considering an automorphism group: the elements that are fixed by every possible automorphism form a field. This leads naturally to the definition of a *fixed field*:

Definition 2.3. (Fixed Field) If H is a subgroup of $Aut(K)$, the group of automorphisms of K, then the subfield F of K that is fixed by all elements of H is called the fixed field of H.

Example 2.2. (Fixed fields) We will use the same examples as **Example 2.1** about automorphism groups:

1. The fixed field of $Aut(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$ is $\mathbb{Q}$. A fixed field must have every element fixed by every possible automorphism.

2. The fixed field of $Aut(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q})$ is $Aut(\mathbb{Q}(\sqrt[3]{2}))$ since the only automorphism is the identity.

In this section so far we have defined all the tools we need to stitch groups and fields together – field create groups by automorphism, and groups create fields by considering fixed fields. Keep these definitions in mind: we are accelerating!

3 Getting Galois

In this section we prove some results concerning the relationship between groups and fields using the definitions from Section 2. 'A few results' may be an understatement – the upcoming propositions are central to piecing together the proof of the Galois Correspondence, where everything will start making sense. So fear not if the motivation behind these results seem unclear on a first read: come back when you completely understand everything in Section 4 and I promise you it's going to look a lot prettier. (At least, I hope that you don't read this handout only once.)

Note on Notation. In order to keep the proofs short I have chosen not to define every single algebraic object in this section (and in the following sections). A general rule of thumb is that E, G, H are groups and F, K, L are fields unless otherwise specified. We're also going to use $E \leq F$ to denote 'E is a subgroup of F' and $F \subseteq K$ to denote 'F is a subfield of K',so apologies if you grew up using another notation.

Proposition 3.1. (Inclusion-reversing Correspondence)

1. If $F_1 \subseteq F_2 \subseteq K$, then $Aut(K/F_2) \leq Aut(K/F_1)$.
2. If $H_1 \leq H_2 \leq Aut(K)$ and H_1, H_2 have fixed fields F_1, F_2, then $F_2 \subseteq F_1$.

Proof.

1. For every $h \in Aut(K/F_2)$ fixing F_2, h is also in $Aut(K/F_1)$ since $F_1 \subseteq F_2$, so $Aut(K/F_2 \leq Aut(K/F_1))$.
2. For every element fixed by some $h \in H_2$, the element must be fixed by every $h \in H_1$ since $H_1 \leq H_2$. Therefore $F_2 \subseteq F_1$.

□

Proposition 3.2. (Degree of Extension vs. Order of Group) Let E be the splitting field over F of the polynomial $f(x) \in F[x]$. Then

$$|Aut(E/F)| \leq [E : F].$$

This proof requires a slightly more nuanced argument and is therefore delegated to the Proofs section. (If you're interested: this is one of the few instances in algebra that proof by induction makes an appearance.)

Proposition 3.2 allows us to relate automorphism groups and fixed fields on a numerical level. Now we have the ideas we need to start looking at things named after Galois:

Definition 3.1. (Galois group.) If $f(x)$ is a separable polynomial over F, then the *Galois group* of $f(x)$ over F is the group of automorphisms of the splitting field of $f(x)$ over F. The Galois group of a field extension is denoted $Gal(K/F)$.

If you're wondering where the separability condition comes from:

Definition 3.2. (Galois extension.) A field extension K/F is defined to be Galois if it is:

1. **(1)** Normal: if $x \in K$ has minimial polynomial $m(x) \in F[x]$, then every root of $m(x)$ is in K.

2. **(2)** Separable: if $x \in K$ has minimial polynomial $m(x) \in F[x]$, then $m(x)$ has distinct roots in its splitting field.

Relationship between Galois group and Galois extension: the automorphism group of K/F is Galois if and only if K/F is a Galois extension. This is by definition.

Note. We've introduced the definition of the normality and separability of field extensions here. If this bothers you, simply acknowledge that in Galois Theory we usually only care about separable, irreducible (and sometimes minimal) polynomials. If you're wondering why a Galois extension is defined this way, it's a little secret :)

Things start to get slightly abstract around here so we introduce some examples. Not everything may be spelled out from now on, so if you don't understand where a statement came from you can think about it in terms of definitions.

Example 3.1. (Galois Groups & Extensions)

1. The extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is Galois (why?) with Galois group isomorphic to $\mathbb{Z}/2\mathbb{Z}$:

$$1 : a + b\sqrt{2} \mapsto a + b\sqrt{2}$$
$$\sigma : a + b\sqrt{2} \mapsto a - b\sqrt{2}$$

2. $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not Galois since $\left|Aut(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q})\right| = 1$. $\mathbb{Q}(\sqrt[3]{2})$ is also not the splitting field of any polynomial. (If you're considering $x^3 - 2$: it has splitting field $\mathbb{Q}(\sqrt[3]{2}, \omega)$ where ω is a primitive third root of unity. More on roots of unity in our next handout.)

3. $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is Galois since it is the splitting field of the separable polynomial $(x^2 - 2)(x^2 - 3)$. The elements in its automorphism group (a Galois group) are:

$$1 : \sqrt{2} \mapsto \sqrt{2},\ \sqrt{3} \mapsto \sqrt{3}$$
$$\sigma : \sqrt{2} \mapsto -\sqrt{2},\ \sqrt{3} \mapsto \sqrt{3}$$
$$\tau : \sqrt{2} \mapsto \sqrt{2},\ \sqrt{3} \mapsto -\sqrt{3}$$
$$\sigma\tau : \sqrt{2} \mapsto -\sqrt{2},\ \sqrt{3} \mapsto -\sqrt{3}$$

which gives the automorphism group as the Klein-4 group.

4 The Galois Correspondence

I will give you a peek:

$$
\begin{array}{ccc}
K & \text{———} & 1 \\
| & & | \\
F_1 & \text{———} & G_i \\
| & & | \\
F_2 & \text{———} & G_{i-1} \\
| & & | \\
\dots & \text{———} & \dots \\
| & & | \\
F_i & \text{———} & G_1 \\
| & & | \\
F & \text{———} & H
\end{array}
$$

No reaction? Here's the complete version:

Theorem 4.1. (Fundamental Theorem of Galois Theory) Let K/F be a Galois extension and set $G = Gal(K/F)$. Then there is a bijection:

$$
\begin{array}{ccc}
K & \text{———} & 1 \\
| & & | \\
F_1 & \text{———} & H_i \\
| & & | \\
\dots & \text{———} & \dots \\
| & & | \\
F & \text{———} & G
\end{array}
$$

Figure 1: subfields of $K \mapsto$ subgroups of G in reverse-inclusion

Under this correspondence:

1. This correspondence is inclusion-reversing, i.e if $E_1 \to H_1$, $E_2 \to H_2$, then $E_1 \subseteq E_2 \Leftrightarrow H_2 \leq H_1$.
2. $[K : E_k] = H_k$ and $[E_k : F] =\mid G : H_k \mid$
3. K/E_k, the intermediate fields over F, is always Galois, with $Gal(K/E_k) = H_k$.
4. E_k/F is Galois if and only if $H_k \leq G$ is a normal subgroup.
5. If E_1, E_2 correspond to H_1, H_2, then $E_1 \bigcap E_2$ corresponds to $\rangle H_1, H_2 \langle$ and the composite field E_1E_2 corresponds to $H_1 \bigcap H_2$.

This correspondence between subfields of K and subgroups of $Gal(K/F)$ is called the *Galois Correspondence.*

Note. The proof of the Fundamental Theorem of Galois Theory is a must-read for everyone; however, due to length concerns of the main section of the handout, it is in the Proofs section. Please check out some of the other proofs that have faded to obscurity down here.